



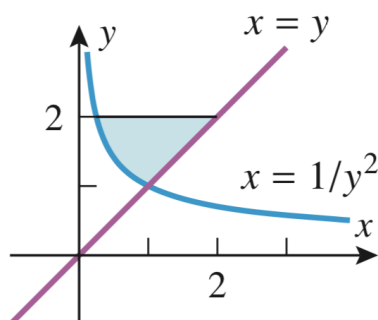
A

Question and Answer Sheet
MTM1502 CALCULUS 2 (Eng.) Gr.2 Midterm Exam 1
(14/04/2026)

Name and Surname

Student Number

QUESTIONS (GROUP A)

Q1)

Which of the following definite integrals is (are) equal to the **area** of the **shaded** region?

I. $\int_1^2 \left(y - \frac{1}{y^2} \right) dy$

II. $\int_{1/4}^1 \left(2 - \frac{1}{\sqrt{x}} \right) dx + \int_1^2 \left(2 - \frac{1}{\sqrt{x}} \right) dx$

III. $\int_{1/4}^1 \left(2 - \frac{1}{\sqrt{x}} \right) dx + \int_1^2 (2 - x) dx$

A) None B) I only C) I and II only D) II only E) **I and III only**

Q2) If the symbol $\lfloor \cdot \rfloor$ denotes the greatest integer function defined by

$$\lfloor x \rfloor = \max \{ n \in \mathbb{Z} \mid n \leq x, x \in \mathbb{R} \},$$

find the *total area* of the region enclosed by $f(x) = x^2 \lfloor 2x \rfloor$, $x = -\frac{1}{2}$, $x = 1$ and the x -axis.

A) $-\frac{1}{3}$ B) $-\frac{1}{4}$ C) 0 D) $\frac{1}{4}$ E) **$\frac{1}{3}$**



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Q3)

$$\int_1^x \frac{f(t)}{t} dt = e^x - a(x-1) - e, \quad x > 0,$$

where f is continuous on $(0, \infty)$. If $\int_1^2 f(x) dx = e^2 - \frac{3}{2}$, then $a = ?$

- A) 0 **B) 1** C) $e-1$ D) e E) $e+1$

Q4) Determine which of the following definite integrals equals $\lim_{n \rightarrow \infty} \sum_{i=1}^n \frac{2n}{n^2 + 4i^2}$.

- A) $\int_0^1 \frac{1}{1+4x^2} dx$ **B) $\int_0^2 \frac{1}{1+x^2} dx$** C) $\int_0^1 \frac{1}{1+x^2} dx$ D) $\int_0^2 \frac{1}{4+x^2} dx$ E) $\int_0^2 \frac{1}{1+2x^2} dx$

Q5) Determine which of the following limits is equal to $\int_0^{\pi/4} \cos(2x) dx$.

- A) **$\lim_{n \rightarrow \infty} \sum_{i=1}^n \frac{\pi}{4n} \cos\left(\frac{\pi i}{2n}\right)$**
B) $\lim_{n \rightarrow \infty} \sum_{i=1}^n \frac{\pi}{2n} \cos\left(\frac{\pi i}{2n}\right)$
C) $\lim_{n \rightarrow \infty} \sum_{i=1}^n \frac{\pi}{4n} \cos\left(\frac{\pi i}{n}\right)$
D) $\lim_{n \rightarrow \infty} \sum_{i=1}^n \frac{1}{n} \cos\left(\frac{\pi i}{2n}\right)$
E) $\lim_{n \rightarrow \infty} \sum_{i=1}^n \frac{\pi}{4n} \cos\left(\frac{\pi i}{4n}\right)$



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Q6) Evaluate the integral $\int \frac{(x + \ln x)^2}{x} dx$.

A) $\frac{1}{2}e^{2x} + \frac{1}{3}x^3 + 2(x-1)e^x + C$

B) $\frac{1}{2}\ln x + \frac{1}{3}e^{3x} + \ln^2 x - 1 + C$

C) $\frac{1}{2}x^2 + \frac{1}{3}\ln^3 x + 2(\ln x - 1)x + C$

D) $\frac{1}{2}x^3 + \frac{1}{3}\ln^2 x + x \ln x + C$

E) $x + e^x + (x-1)\ln x + C$

Q7) Evaluate the integral $\int \frac{x^2 + 4x - 1}{x^4 - 1} dx$.

A) $\ln|x^2 - 1| + \arctan x + C$

B) $\ln\left|\frac{x-1}{x+1}\right| + \arctan x + C$

C) $\ln|x^2 - 1| + \arcsin x + C$

D) $\ln\left|\frac{x^2 - 1}{x^2 + 1}\right| + \arctan x + C$

E) $\ln\left|\frac{x}{x^2 + 1}\right| + \arctan x + C$

Q8) Evaluate the integral $\int \frac{dx}{\sqrt{1 - e^x}}$.

A) $x \ln \sqrt{\frac{1+e^x}{1-e^x}} + C$

B) $\sqrt{1+e^x} + \ln|1-e^x| + C$

C) $\sqrt{1-e^x} + \ln|1+e^x| + C$

D) $\ln \sqrt{\frac{1-e^x}{1+e^x}} + C$

E) $\ln \left| \frac{1 - \sqrt{1 - e^x}}{1 + \sqrt{1 - e^x}} \right| + C$



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Q9) Evaluate the integral $\int \frac{x^2 \arctan x}{1+x^2} dx$.

A) $x \arctan x - \frac{1}{2} \ln(1+x^2) + C$

B) $x \arctan x - \frac{1}{2} \ln(1+x^2) - \frac{1}{2} \arctan^2 x + C$

C) $\arctan x - x \ln(1+x^2) + C$

D) $\arctan x - \frac{1}{2} \ln(1+x^2) - \frac{1}{2} x^2 + C$

E) $\arctan(1+x^2) - \frac{1}{2} \ln(1+x^2) - \frac{1}{2} x^2 + C$

Q10) Evaluate the integral $\int \frac{dx}{x^4 \sqrt{1-x^2}}$.

A) $-\frac{\sqrt{1-x^2}}{x} - \frac{1}{3} \frac{\sqrt{(1-x^2)^3}}{x^3} + C$

B) $-\cot x - \frac{1}{3} \cot^3 x + C$

C) $-\frac{x}{1-x^2} + \frac{x^3}{\sqrt{(1-x^2)^3}} + C$

D) $-\frac{x}{1-x^2} - \frac{x^3}{\sqrt{1-x^2}} + C$

E) $\tan x - \frac{1}{3} \tan^3 x + C$



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Q11) For which values of p does the integral

$$\int_1^{\infty} \frac{\sqrt{x} + \sin x}{x^p} dx$$

converge?

A) $p < 1$

B) $p > \frac{3}{2}$

C) $p > 2$

D) $p < 2$

E) $p < \frac{3}{2}$

Q12) Determine whether the integral

$$\int_0^{\infty} \frac{\arctan x}{1 + x^2} dx$$

is convergent or divergent.

A) It is divergent.

B) It is convergent, and its value is $\frac{\pi}{4}$.

C) It is convergent, and its value is $\frac{\pi^2}{8}$.

D) It is convergent, and its value is $\frac{\pi}{2}$.

E) It is convergent, and its value is $\frac{\pi^2}{4}$.

Q13) Determine whether the integral

$$\int_0^{\pi} \frac{\sin x}{\cos^2 x} dx$$

is convergent or divergent.

A) It is divergent.

B) It is convergent, and its value is $\frac{\pi}{2}$.

C) It is convergent, and its value is 0.

D) It is convergent, and its value is π .

E) It is convergent, and its value is 1.



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Q14) Determine whether the integral

$$\int_{-\infty}^{\infty} \frac{1}{x^2 + 2x + 5} dx$$

is convergent or divergent.

A) It is divergent.

B) It is convergent, and its value is $\frac{\pi}{2}$.

C) It is convergent, and its value is $\frac{\pi}{4}$.

D) It is convergent, and its value is $\frac{\pi}{8}$.

E) It is convergent, and its value is π .

Q15) Determine whether the integral

$$\int_0^{e^2} \frac{1}{x(1 - \ln x)} dx$$

is convergent or divergent.

A) It is divergent.

B) It is convergent, and its value is 0.

C) It is convergent, and its value is e .

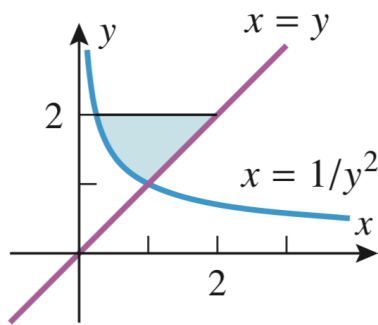
D) It is convergent, and its value is 2.

E) It is convergent, and its value is e^2 .

	SUGGESTED QUESTIONS (B. Ali İBRAHİMOĞLU) MTM1502 CALCULUS 2 RESIT EXAM (14/04/2026)		

QUESTIONS AND SOLUTIONS

Q1)



Which of the following definite integrals is (are) equal to the **area** of the **shaded** region?

- I. $\int_1^2 \left(y - \frac{1}{y^2} \right) dy$
- II. $\int_{1/4}^1 \left(2 - \frac{1}{\sqrt{x}} \right) dx + \int_1^2 \left(2 - \frac{1}{\sqrt{x}} \right) dx$
- III. $\int_{1/4}^1 \left(2 - \frac{1}{\sqrt{x}} \right) dx + \int_1^2 (2 - x) dx$

- A) None B) I only C) I and II only
- D) II only E) I and III only

Solution of Q1) Answer: E)

Integrating with respect to y :

$$\begin{aligned} \text{Area} &= \int_1^2 \left(y - \frac{1}{y^2} \right) dy \\ &= \left(\frac{y^2}{2} + \frac{1}{y} \right) \Big|_1^2 \\ &= 1 \end{aligned}$$

Integrating with respect to x :

$$\text{Area} = \int_{1/4}^1 \left(2 - \frac{1}{\sqrt{x}} \right) dx + \int_1^2 (2 - x) dx = 1$$

Q2)

If the symbol $\lfloor \cdot \rfloor$ denotes the greatest integer function defined by

$$\lfloor x \rfloor = \max \{ n \in \mathbb{Z} \mid n \leq x, x \in \mathbb{R} \},$$

find the *total area* of the region enclosed by

$$f(x) = x^2 \lfloor 2x \rfloor, \quad x = -\frac{1}{2}, \quad x = 1 \text{ and the } x\text{-axis.}$$

- A) $-\frac{1}{3}$ B) $-\frac{1}{4}$ C) 0 D) $\frac{1}{4}$ E) $\frac{1}{3}$

Solution of Q2) Answer: E) 1/3

$$\lfloor 2x \rfloor = \begin{cases} -1, & -0.5 \leq x < 0 \Rightarrow -1 \leq 2x < 0 \\ 0, & 0 \leq x < 0.5 \Rightarrow 0 \leq 2x < 1 \\ 1, & 0.5 \leq x < 1 \Rightarrow 1 \leq 2x < 2 \end{cases}$$

$$\Rightarrow x^2 \lfloor 2x \rfloor = \begin{cases} -x^2, & -0.5 \leq x < 0 \\ 0, & 0 \leq x < 0.5 \\ x^2, & 0.5 \leq x < 1 \end{cases}$$

$$\text{total area} = \int_{-1/2}^1 |x^2 \lfloor 2x \rfloor| dx = \int_{-1/2}^0 -x^2 dx + \int_0^{1/2} 0 dx + \int_{1/2}^1 x^2 dx$$

$$= \frac{1}{3}$$

	SUGGESTED QUESTIONS (B. Ali İBRAHİMOĞLU) MTM1502 CALCULUS 2 RESIT EXAM (14/04/2026)		

Q3)

$$\int_1^x \frac{f(t)}{t} dt = e^x - a(x-1) - e, \quad x > 0,$$

where f is continuous on $(0, \infty)$. If

$$\int_1^2 f(x) dx = e^2 - \frac{3}{2}, \text{ then } a = ?$$

- A) 0 B) 1 C) $e-1$ D) e E) $e+1$

Solution of Q3) Answer: B) 1

Using the FTC Part1:

$$\frac{d}{dx} \left(\int_1^x \frac{f(t)}{t} dt \right) = \frac{f(x)}{x},$$

and differentiate the right-hand side:

$$\frac{d}{dx} (e^x - a(x-1) - e) = e^x - a.$$

Hence, $\frac{f(x)}{x} = e^x - a$. So, $f(x) = xe^x - ax$.

$$\int_1^2 (xe^x - ax) dx = e^2 - \frac{3}{2}$$

$$\Rightarrow \left[(x-1)e^x - \frac{ax^2}{2} \right]_{x=1}^{x=2} = e^2 - \frac{3a}{2} = e^2 - \frac{3}{2} \Rightarrow \boxed{a=1}$$

Q4)

Determine which of the following definite integrals

equals $\lim_{n \rightarrow \infty} \sum_{i=1}^n \frac{2n}{n^2 + 4i^2}$.

- A) $\int_0^1 \frac{1}{1+4x^2} dx$ B) $\int_0^2 \frac{1}{1+x^2} dx$ C) $\int_0^1 \frac{1}{1+x^2} dx$
D) $\int_0^2 \frac{1}{4+x^2} dx$ E) $\int_0^2 \frac{1}{1+2x^2} dx$

Solution of Q4) Answer: B) $\int_0^2 \frac{1}{1+x^2} dx$

$$\frac{2n}{n^2 + 4i^2} = \frac{2}{n} \cdot \frac{1}{1 + (2i/n)^2}.$$

Thus, this is a Riemann sum with

$$\Delta x = \frac{2}{n}, \quad x_i = \frac{2i}{n},$$

$$\text{So, } \lim_{n \rightarrow \infty} \sum_{i=1}^n \frac{2n}{n^2 + 4i^2} = \lim_{n \rightarrow \infty} \sum_{i=1}^n \underbrace{\frac{1}{1 + (2i/n)^2}}_{f(x_i)} \underbrace{\frac{2}{n}}_{\Delta x} = \int_0^2 \frac{1}{1+x^2} dx.$$

Q5)

Determine which of the following limits is equal to

$$\int_0^{\pi/4} \cos(2x) dx.$$

- A) $\lim_{n \rightarrow \infty} \sum_{i=1}^n \frac{\pi}{4n} \cos\left(\frac{\pi i}{2n}\right)$ B) $\lim_{n \rightarrow \infty} \sum_{i=1}^n \frac{\pi}{2n} \cos\left(\frac{\pi i}{2n}\right)$
C) $\lim_{n \rightarrow \infty} \sum_{i=1}^n \frac{\pi}{4n} \cos\left(\frac{\pi i}{n}\right)$ D) $\lim_{n \rightarrow \infty} \sum_{i=1}^n \frac{1}{n} \cos\left(\frac{\pi i}{2n}\right)$
E) $\lim_{n \rightarrow \infty} \sum_{i=1}^n \frac{\pi}{4n} \cos\left(\frac{\pi i}{4n}\right)$

Solution of Q5) Answer: A) $\lim_{n \rightarrow \infty} \sum_{i=1}^n \frac{\pi}{4n} \cos\left(\frac{\pi i}{2n}\right)$

$$\Delta x = \frac{\pi/4 - 0}{n} = \frac{\pi}{4n}, \quad x_i = \frac{\pi i}{4n}.$$

$$\text{So, } \cos(2x_i) = \cos\left(2 \cdot \frac{\pi i}{4n}\right) = \cos\left(\frac{\pi i}{2n}\right).$$

$$\text{Therefore, } \int_0^{\pi/4} \cos(2x) dx = \lim_{n \rightarrow \infty} \sum_{i=1}^n \underbrace{\cos\left(\frac{\pi i}{2n}\right)}_{f(x_i)} \underbrace{\frac{\pi}{4n}}_{\Delta x}.$$

	YTÜ – Kimya-Metalurji Fakültesi, Soru ve Cevap Kağıdı	NOT TABLOSU				
		S1	S2	S3	S4	Toplam
Öğrenci Adı ve Soyadı						
Öğrenci Numarası						
Bölüm	Matematik Mühendisliği				Sınav Tarihi	14 / 04 / 2026
Ders Adı	MTM1502 Analiz II I. VİZE Sınavı (Bahar 25-26)	Grup Numarası	3	Sınav Süresi	...	Sınıf
Dersi Veren Adı ve Soyadı					dk.	☐ KMB ☐ KMB
					İmza	
YÖK nun 2547 sayılı Kanunun Öğrenci Disiplin Yönetmeliğinin 9. Maddesi olan “Sınavlarda kopya yapmak ve yaptırmak veya buna teşebbüs etmek” fiili işleyenler bir veya iki yarıyıl uzaklaştırma cezası alırlar.						

Soru 1. $\int \frac{(x + \ln x)^2}{x} dx$ integralini hesaplayınız. a) $\frac{1}{2}e^{2x} + \frac{1}{3}x^3 + 2(x-1)e^x + C$ b) $\frac{1}{2} \ln x + \frac{1}{3}e^{3x} + \ln^2 x - 1 + C$ c) $\frac{1}{2}x^2 + \frac{1}{3}\ln^3 x + 2(\ln x - 1)x + C$ d) $\frac{1}{2}x^3 + \frac{1}{3}\ln^2 x + x \ln x + C$ e) $x + e^x + (x-1) \ln x + C$	$\ln x = y, \frac{1}{x} dx = dy$ $I = \int \frac{(x + \ln x)^2}{x} dx = \int (e^y + y)^2 dy = \int e^{2y} dy + 2 \int ye^y dy + \int y^2 dy$ $I = \frac{e^{2y}}{2} + \frac{y^3}{3} + 2 \int ye^y dy$ $y = u \quad e^y dy = dv$ $dy = du \quad e^y = v$ $I = \frac{e^{2y}}{2} + \frac{y^3}{3} + 2(ye^y - \int e^y dy) = \frac{e^{2y}}{2} + \frac{y^3}{3} + 2(y-1)e^y + C$ $= \frac{1}{2}x^2 + \frac{1}{3}\ln^3 x + 2(\ln x - 1)x + C$ Cevap C
Soru 2. $\int \frac{x^2 + 4x - 1}{x^4 - 1} dx$ integralini hesaplayınız. a) $\ln x^2 - 1 + \arctan x + C$ b) $\ln \left \frac{x-1}{x+1} \right + \arctan x + C$ c) $\ln x^2 - 1 + \arcsin x + C$ d) $\ln \left \frac{x^2 - 1}{x^2 + 1} \right + \arctan x + C$ e) $\ln \left \frac{x}{x^2 + 1} \right + \arctan x + C$	$\frac{x^2 + 4x - 1}{x^4 - 1} = \frac{x^2 + 4x - 1}{(x-1)(x+1)(x^2 + 1)} = \frac{A}{x-1} + \frac{B}{x+1} + \frac{Cx + D}{x^2 + 1}$ $= \frac{(A+B+C)x^3 + (A-B+D)x^2 + (A+B-C)x + A-B-D}{(x-1)(x+1)(x^2 + 1)}$ $\left. \begin{aligned} A+B+C &= 0 \\ A-B+D &= 1 \\ A+B-C &= 4 \\ A-B-D &= -1 \end{aligned} \right\} C = -2, D = 1, A = 1, B = 1$ $\int \frac{x^2 + 4x - 1}{x^4 - 1} dx = \int \frac{1}{x-1} dx + \int \frac{1}{x+1} dx - \int \left(\frac{2x}{x^2 + 1} - \frac{1}{x^2 + 1} \right) dx$ $= \ln \left \frac{x^2 - 1}{x^2 + 1} \right + \arctan x + C$ Cevap D
Soru 3. $\int \frac{dx}{\sqrt{1-e^x}}$ integralini hesaplayınız. a) $x \ln \sqrt{\frac{1+e^x}{1-e^x}} + C$ b) $\sqrt{1+e^x} + \ln 1-e^x + C$ c) $\sqrt{1-e^x} + \ln 1+e^x + C$	$\sqrt{1-e^x} = u, -\frac{e^x}{2\sqrt{1-e^x}} dx = du \rightarrow dx = -\frac{2u}{1-u^2} du$ $\int \frac{dx}{\sqrt{1-e^x}} = -\int \frac{1}{u} \frac{2u}{1-u^2} du = -2 \int \frac{1}{1-u^2} du$ $= -\int \left(\frac{1}{1-u} + \frac{1}{1+u} \right) du = \ln \left \frac{1-u}{1+u} \right + C = \ln \left \frac{1-\sqrt{1-e^x}}{1+\sqrt{1-e^x}} \right + C$ Cevap E

Başarılar.

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Öğrenci Adı ve Soyadı						
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Bölüm	Matematik Mühendisliği			Sınav Tarihi	14 / 04 / 2026	
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Dersi Veren Adı ve Soyadı					dk.	☐ KMB
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d) $\ln \sqrt{\frac{1-e^x}{1+e^x}} + C$ e) $\ln \left \frac{1-\sqrt{1-e^x}}{1+\sqrt{1-e^x}} \right + C$	
Soru 4. $\int \frac{x^2 \arctan x}{1+x^2} dx$ integralini hesaplayınız. a) $x \arctan x - \frac{1}{2} \ln(1+x^2) + C$ b) $x \arctan x - \frac{1}{2} \ln(1+x^2) - \frac{1}{2} \arctan^2 x + C$ c) $\arctan x - x \ln(1+x^2) + C$ d) $\arctan x - \frac{1}{2} \ln(1+x^2) - \frac{1}{2} x^2 + C$ e) $\arctan(1+x^2) - \frac{1}{2} \ln(1+x^2) - \frac{1}{2} x^2 + C$	$\arctan x = y, \frac{1}{1+x^2} dx = dy$ $I = \int \frac{x^2 \arctan x}{1+x^2} dx = \int y \tan^2 y dy = \int y(1 + \tan^2 y) dy - \int y dy$ $I = \int y(1 + \tan^2 y) dy - \frac{y^2}{2}$ $y = u \quad (1 + \tan^2 y) dy = dv$ $dy = du \quad \tan y = v$ $I = y \tan y - \int \tan y dy - \frac{y^2}{2} = y \tan y + \ln \cos y - \frac{y^2}{2} + C$ $= x \arctan x - \frac{1}{2} \ln(1+x^2) - \frac{1}{2} \arctan^2 x + C$ Cevap B
Soru 5. $\int \frac{dx}{x^4 \sqrt{1-x^2}}$ integralini hesaplayınız. a) $-\frac{\sqrt{1-x^2}}{x} - \frac{1}{3} \frac{\sqrt{(1-x^2)^3}}{x^3} + C$ b) $-\cot x - \frac{1}{3} \cot^3 x + C$ c) $-\frac{x}{1-x^2} + \frac{x^3}{\sqrt{(1-x^2)^3}} + C$ d) $-\frac{x}{1-x^2} - \frac{x^3}{\sqrt{1-x^2}} + C$ e) $\tan x - \frac{1}{3} \tan^3 x + C$	$x = \sin u, dx = \cos u du$ $I = \int \frac{dx}{x^4 \sqrt{1-x^2}} = \int \frac{\cos u du}{\sin^4 u \cos u} = \int \frac{du}{\sin^4 u}$ $= \int \frac{1}{\sin^2 u} \frac{1}{\sin^2 u} du = \int (1 + \cot^2 u) \frac{1}{\sin^2 u} du$ $z = \cot u, dz = -\frac{1}{\sin^2 u} du$ $I = -\int (1 + z^2) dz = -z - \frac{1}{3} z^3 + C = -\cot u - \frac{1}{3} (\cot u)^3 + C$ $= -\frac{\sqrt{1-x^2}}{x} - \frac{1}{3} \frac{\sqrt{(1-x^2)^3}}{x^3} + C$ Cevap A

Başarılar.

SORULAR

1)

$$\int_1^{\infty} \frac{\sqrt{x} + \sin x}{x^p} dx$$

integralinin yakınsak olması için p değeri hangi aralıkta olmalıdır?

- A) $p < 1$
- B) $p > \frac{3}{2}$**
- C) $p > 2$
- D) $p < 2$
- E) $p < \frac{3}{2}$

Çözüm 1)

Doğrudan karşılaştırma testi uygulanarak

$$\frac{\sqrt{x} + \sin x}{x^p} \leq \frac{\sqrt{x} + 1}{x^p}$$

$$\frac{\sqrt{x}}{x^p} + \frac{1}{x^p}$$

$p > 1$ olmalı

$$\frac{1}{x^{p-\frac{1}{2}}}$$

$p - \frac{1}{2} > 1$ olmalı

Cevap: B

$p > \frac{3}{2}$

2)

$$\int_0^{\infty} \frac{\arctan x}{1+x^2} dx$$

integralinin yakınsaklığı/ıraksaklığını inceleyiniz.

- A) İraksaktır.
- B) Yakınsaktır, yakınsadığı değer $\frac{\pi}{4}$ dür.
- C) Yakınsaktır, yakınsadığı değer $\frac{\pi^2}{8}$ dir.**
- D) Yakınsaktır, yakınsadığı değer $\frac{\pi}{2}$ dir.
- E) Yakınsaktır, yakınsadığı değer $\frac{\pi^2}{4}$ dür.

Limit Karşılaştırma testi kullanılarak

$$f(x) = \frac{\arctan x}{1+x^2}, \quad g(x) = \frac{1}{1+x^2}$$

$$\lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} = \lim_{x \rightarrow \infty} \arctan x = \frac{\pi}{2} = L \text{ (sönü)}$$

$$\frac{1}{1+x^2} < \frac{1}{x^2} \Rightarrow \text{Doğrudan kos. testinden } \Rightarrow \text{yakınsak}$$

$p=2 > 1$ için yakınsak

Çözüm 2)

$$\arctan x = u \Rightarrow \frac{1}{1+x^2} dx = du$$

değişken

$$\text{değiştime ile } \int u du = \frac{u^2}{2} + C$$

$$\int_0^{\infty} \frac{\arctan x}{1+x^2} dx = \lim_{b \rightarrow \infty} \int_0^b \frac{\arctan x}{1+x^2} dx$$

$$= \lim_{b \rightarrow \infty} \left[\frac{(\arctan x)^2}{2} \right]_0^b$$

$$= \lim_{b \rightarrow \infty} \left[\frac{(\arctan b)^2}{2} - \frac{(\arctan 0)^2}{2} \right]$$

$$= \left(\frac{\pi}{2} \right)^2 / 2 - 0 = \frac{\pi^2}{8}$$

$$\int_0^{\infty} \frac{\arctan x}{1+x^2} dx \text{ yakınsak}$$

Cevap: C

3)

$$\int_0^{\pi} \frac{\sin x}{\cos^2 x} dx$$

integralinin yakınsaklığı/ıraksaklığını inceleyiniz.

- A) ıraksaktır.
 B) Yakınsaktır, yakınsadığı değer $\frac{\pi}{2}$ dir.
 C) Yakınsaktır, yakınsadığı değer 0 dir.
 D) Yakınsaktır, yakınsadığı değer π dir.
 E) Yakınsaktır, yakınsadığı değer 1 dir.

Çözüm 3)

$$f(x) = \frac{\sin x}{\cos^2 x} \Rightarrow \cos^2 x = 0 \Rightarrow \cos x = 0 \Rightarrow x = \frac{\pi}{2}$$

süreksizlik

$$\int_0^{\pi} \frac{\sin x}{\cos^2 x} dx = \int_0^{\frac{\pi}{2}} \frac{\sin x}{\cos^2 x} dx + \int_{\frac{\pi}{2}}^{\pi} \frac{\sin x}{\cos^2 x} dx \quad \left[\begin{array}{l} u = \cos x \\ du = -\sin x \end{array} \right]$$

$$\lim_{b \rightarrow \frac{\pi}{2}^-} \int_0^b \frac{\sin x}{\cos^2 x} dx = \lim_{b \rightarrow \frac{\pi}{2}^-} \left[-\frac{1}{\cos x} \right]_0^b$$

$$= \lim_{b \rightarrow \frac{\pi}{2}^-} \left[-\frac{1}{\cos b} + \frac{1}{\cos 0} \right]$$

$\Rightarrow$ ıraksak
cevap: A

4)

$$\int_{-\infty}^{\infty} \frac{1}{x^2 + 2x + 5} dx$$

integralinin yakınsaklığı/ıraksaklığını inceleyiniz.

- A) ıraksaktır.
 B) Yakınsaktır, yakınsadığı değer $\frac{\pi}{2}$ dir.
 C) Yakınsaktır, yakınsadığı değer $\frac{\pi}{4}$ dir.
 D) Yakınsaktır, yakınsadığı değer $\frac{\pi}{8}$ dir.
 E) Yakınsaktır, yakınsadığı değer π dir.

Çözüm 4)

$$x^2 + 2x + 5 = (x+1)^2 + 4$$

$$\Rightarrow \int_{-\infty}^{\infty} \frac{1}{(x+1)^2 + 4} dx$$

$$= \lim_{a \rightarrow -\infty} \int_a^0 \frac{1}{(x+1)^2 + 4} dx + \lim_{b \rightarrow \infty} \int_0^b \frac{1}{(x+1)^2 + 4} dx$$

$$= \lim_{a \rightarrow -\infty} \frac{1}{2} \arctan \left(\frac{x+1}{2} \right) \Big|_a^0 + \lim_{b \rightarrow \infty} \frac{1}{2} \arctan \left(\frac{x+1}{2} \right) \Big|_0^b$$

$$= \lim_{a \rightarrow -\infty} \frac{1}{2} \arctan \left(\frac{a+1}{2} \right) + \lim_{b \rightarrow \infty} \frac{1}{2} \arctan \left(\frac{b+1}{2} \right)$$

$$= \frac{\pi}{4} + \frac{\pi}{4} = \frac{\pi}{2} \quad \text{cevap: B}$$

5)

$$\int_0^{e^2} \frac{1}{x(1-\ln x)} dx$$

integralinin yakınsaklığı/ıraksaklığını inceleyiniz.

- A) ıraksaktır.
 B) Yakınsaktır, yakınsadığı değer 0 dir.
 C) Yakınsaktır, yakınsadığı değer e dir.
 D) Yakınsaktır, yakınsadığı değer 2 dir.
 E) Yakınsaktır, yakınsadığı değer e^2 dir.

Çözüm 5)

$$x(1-\ln x) = 0 \Rightarrow \begin{array}{l} x=0 \\ \vee \\ 1-\ln x=0 \Rightarrow \ln x=1 \Rightarrow x=e \end{array}$$

nok. da süreksizlik var.

$$\int_0^{e^2} \frac{dx}{x(1-\ln x)} = \int_0^e \frac{dx}{x(1-\ln x)} + \int_e^{e^2} \frac{dx}{x(1-\ln x)}$$

$$\lim_{a \rightarrow e^+} \int_a^{e^2} \frac{dx}{x(1-\ln x)} = \lim_{a \rightarrow e^+} \left[-\ln |1-\ln x| \right]$$

$$= \lim_{a \rightarrow e^+} \left[-\ln |1-\ln(e^2)| \right]$$

$$= \lim_{a \rightarrow e^+} \left[-\ln(1-2) - (-\ln(1-\ln a)) \right]$$

$\Rightarrow$ ıraksaktır. Cevap: A

